

Andrew Park

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RESEARCH INTEREST

Heterogeneous Architecture and **Heterogeneous Computing** for emerging workloads (i.e. artificial intelligence),
Software Optimization for heterogeneous computing

EDUCATION

Yonsei University, Seoul, Republic of Korea

Expected **August 2025**

BS, Electrical and Electronic Engineering

Thesis: “Limitation of Optimization Targets on Loop Optimizers”

- Advisor: Prof. Hanjun Kim
- Relevant Coursework: Computer Architecture, Microprocessors, System IC Design

RESEARCH EXPERIENCE

Yonsei University, Seoul, Republic of Korea

March – June 2024

Undergraduate Thesis Student

Advisor: Prof. Hanjun Kim

Thesis: “Limitation of Optimization Targets on Loop Optimizers”

- Identified LLVM’s polyhedral optimizer Polly’s limitations in optimizing codes with dynamically bounded loops
- Restructured Polly’s target-setting modules to uncover all loops accurately for 3.5x faster optimization results

Seoul National University, Seoul, Republic of Korea

June – August 2024

Undergraduate Research Assistant

Project: Development of near-data computing solution targeting MoE LLMs.

Advisor: Prof. Hyunkjae Lee

- Studied mixture-of-experts models’ functionality and characteristics
- Implemented a near-data computing solution that alleviates bottlenecks between system memory and device using Verilog on AMD/Xilinx U280 FPGA
- Achieved 2.7x gain for ‘google/switch-base-8’ summarization task

PROJECTS

IEEE754 single precision FPU development

May – August 2024

- Studied various types of adders; applied Kogge-Stone adder
- Examined patterns of usage of adders while FPU is performing its functions
- Implemented adder multiplexing scheme to optimize resource utilization
- Implemented control circuit to monitor calculation status and generate quick finish signal

WORK EXPERIENCE

Insight Edu Inc., Seoul, Republic of Korea **Jan 2019 - Nov 2021, Mar 2024 - Present**
Developing team leader for development of learning management system

Seoul National School for the Blind, Seoul, Republic of Korea **Jan 2022 - Oct 2023**
Assistant teacher for visually impaired students

AWARDS

Merit-based scholarship, Yonsei University College of Engineering **Spring, Fall 2024**
Educational Outreach Scholarship, Yonsei University **2019**
Life Academy Scholarship, Yonsei University **2019**

SKILLS AND LANGUAGES

Languages: C, C++, Verilog, MATLAB, Python, Javascript, React

Technical: Xcelium, Vivado, Vitis, v++, Synopsys Design Compiler, Quartus, Pybind, GCC, LLVM, Polly, AWS, GCP, Firebase

General: English (Fluent), Korean (Native)